

UNDAPALLI RAVI KUMAR

ETL & Data Engineer | AWS Cloud | Pipeline Automation

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PROFESSIONAL SUMMARY

ETL & Data Engineer with 4.5+ years of experience building and supporting enterprise-scale production data pipelines for a GE Healthcare engagement. Skilled in Fivetran, HVR, custom Python/SQL/PySpark scripts, AWS Glue, AWS AppFlow, Apache NiFi, and Informatica, along with AWS data services (S3, Redshift, Lambda, Step Functions, EMR) for cloud data engineering and integration. Experienced in production troubleshooting, P1/P2 incident management, RCA, and SLA-driven operations, with additional exposure to on-premises-to-AWS cloud migration. Comfortable working directly with client and cross-functional teams to keep high-volume ingestion workflows stable and reliable.

PROFESSIONAL EXPERIENCE

System Engineer – ETL & Data Engineering

Tata Consultancy Services (TCS) | Dec 2021 – Present

- Build and maintain enterprise-scale production ETL and data ingestion pipelines using Fivetran, HVR, custom Python/SQL/PySpark scripts, AWS Glue, AWS AppFlow, Apache NiFi, and Informatica, supporting a GE Healthcare client engagement.
- Identified and resolved a CDC/deduplication data integrity issue in an AWS AppFlow → S3 → AWS Glue → Staging → Main ingestion pipeline: the existing logic ranked records by an audit date field with a fixed 00:00:00 time component, so same-day updates could not be reliably ordered and stale records risked being promoted as the latest version.
- Redesigned the deduplication logic using a composite ranking key (audit date combined with batch ID / ingestion timestamp) to enforce deterministic ordering of same-day updates, addressing the issue across 200+ tables running on multiple ingestion schedules and addressing the associated data integrity and audit risk.
- Develop and maintain AWS cloud data workflows across S3, Glue, AppFlow, Step Functions, Lambda, EMR, and Redshift for data ingestion, transformation, and downstream reporting.
- Build serverless ETL jobs with AWS Glue and Lambda, using Step Functions for workflow orchestration and Redshift for data warehousing.
- Lead Major Incident Management (MIM) calls and resolve P1/P2 incidents within SLA, producing RCA documentation and follow-up action plans.
- Automate 10+ manual operational workflows using Python, PySpark, and Unix shell scripting, reducing manual effort involved in daily pipeline health monitoring.
- Write SQL, Python, and PySpark scripts for data extraction, job health analysis, and performance monitoring across ingestion workflows to support proactive issue detection.
- Support on-premises-to-AWS cloud migration for multiple applications, working to maintain data integrity and minimal downtime during cutover.
- Manage end-to-end production job lifecycle (initialization, finalization, data load) via ServiceNow (Incident, Change, RITM) following ITIL practices.
- Perform system health monitoring, patch management, memory allocation checks, and vulnerability remediation using Unix shell scripting and remote tools (Putty, WinSCP).
- Work with cross-functional stakeholders to troubleshoot pipeline issues and act as a technical point of contact between client and internal teams.
- Apply foundational knowledge of Google Cloud Platform (Cloud Storage, BigQuery, Dataproc, IAM basics) in cross-cloud data engineering discussions.

TECHNICAL SKILLS

ETL & Data Engineering: Fivetran, HVR, Custom Python/SQL/PySpark Scripts, AWS Glue, AWS AppFlow, Apache NiFi, Informatica, CDC

Cloud Platforms & Services: AWS (S3, Glue, AppFlow, Redshift, Lambda, Step Functions, EMR), Google Cloud Platform – basics (Cloud Storage, BigQuery, Dataproc, IAM, Compute Engine fundamentals), Azure DevOps

Programming & Scripting: Python, SQL, PySpark, Unix/Linux Shell Scripting

Data Operations: Pipeline Monitoring, Job Scheduling, Data Quality Checks, SLA Management, Production Troubleshooting

ITSM & Tools: ServiceNow, ITIL (Incident, Change, Problem, RITM), RCA Documentation, MIM Leadership, Jira, Putty, WinSCP, Linux, Windows Server

CERTIFICATIONS

Fivetran 6.0 Technical Certification

Automated ELT pipelines, connector setup, schema management

Apache NiFi – Automation & ETL

Data flow design, ingestion, transformation, routing

TCS GenAI Foundations (2025)

LLM fundamentals, prompt engineering, responsible AI

TCS GenAI Practitioner (2025)

Applied GenAI workflows for data engineering use cases

AWARDS & RECOGNITION

- Service & Commitment Award, TCS – recognized for consistent performance and reliability in production data engineering operations.
- On-the-Spot Award (Multiple) – for rapid resolution of critical P1/P2 incidents.
- Client & Manager Appreciation – recognized for proactive issue handling and clear technical communication.

EDUCATION

B.Tech – Electronics & Communication Engineering, 2021

Swarnandhra College of Engineering & Technology, JNTUK, Narsapur | CGPA: 7.01

Intermediate (MPC), 2017

Aditya Junior College, Narsapur | 92.2%

SSC (10th Standard), 2015

Z.P.P.H School, L.B Cherla | CGPA: 8.5